



DzNet Signal Coverage

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Difficulty Level : Medium

Description

A new telecom operator in Algeria wants to deploy its network across multiple cities.

The cities are modeled as a **graph**:

- Each **city** is a node
- Connections between cities are represented by an **adjacency matrix**

The company can install a **signal tower** in any city.

A signal tower installed in a city covers:

- The city itself
- All **directly connected neighboring cities**

Note

- Some cities may have **no connections**
- A city can be covered by **multiple towers**

Task: Determine the **minimum** number of **signal towers** required so that every city is covered.

Examples

Input	Output
4 0100 1010 0101 0010	2
6 011000 101000 110000 000011 000101 000110	2
12 011000000000 101100000000 110000000000 010010000000 000101100000 000010100000 000011010000 000000101100 000000010100 000000011011 000000000101 000000000110	3